

PLATFORM OVERVIEW

Every data source, one governed path to the lakehouse

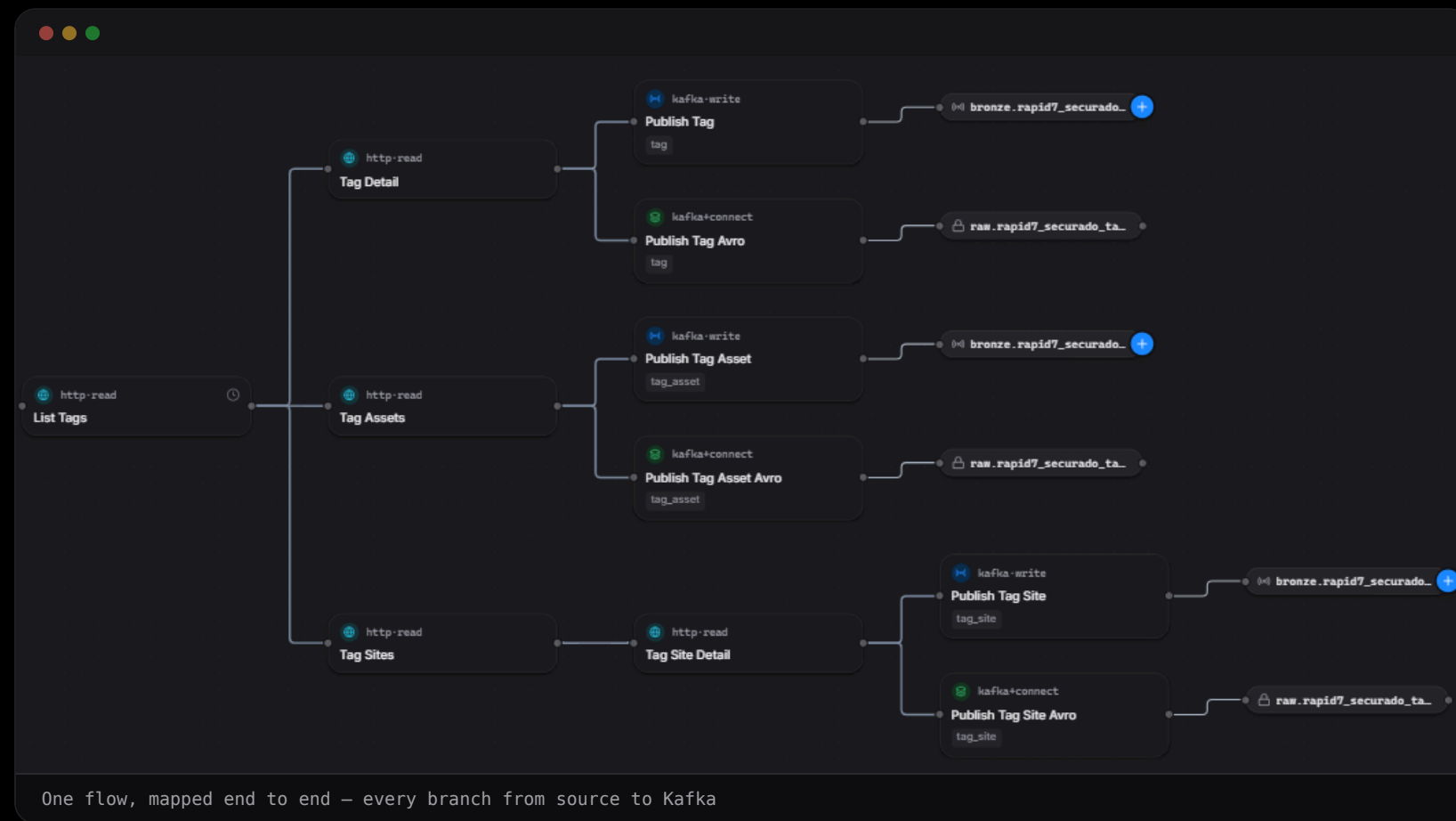
Connect a source, build the pipeline visually, and publish with confidence — without writing a flow by hand or guessing at a schema.

Self-service onboarding

Schema optional, not required

One control plane

Full audit trail



THE PROBLEM

Every new source meant a pipeline built from nothing

There was no shared platform for data ingestion. Each team stood up its own flow, invented its own schema, and hand-wired its own path into the lakehouse.

01

Flows built by hand, one at a time

Every new source — or new data type from a source already connected — meant another flow written from scratch. Complexity multiplied with every addition.

03

Lakehouse sinks wired from memory

Pointing Kafka at the lakehouse meant hand-configuring the sink connector, the topic, and the schema. Get one wrong and it fails silently.

02

Schemas invented on the spot

No registry, no ownership, no version history. Stability meant someone remembering to write a schema by hand — and hoping it still matched reality.

04

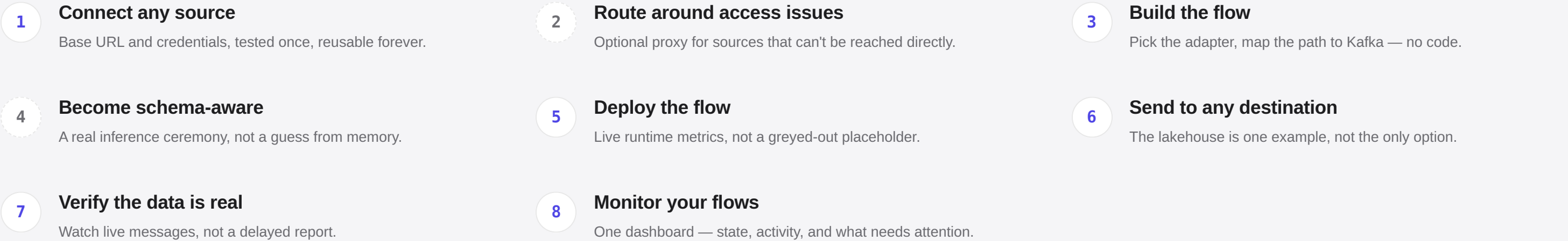
Some sources simply couldn't be reached

APIs without proper certificates had no centralized, safe way to be routed to — with no shared proxy layer to solve it once.

**It shouldn't take a specialist and a week to connect one more source. So
we made it a workflow, not a project.**

From an empty platform to verified data — eight steps

Here's the actual path a team takes the first time they use Data Mobility.



STEP · 1 / 8

Connect any source

The first thing a team does is register the system they want to pull from — a scanner, a SIEM, a database — as an Application Service: base URL and credentials, tested once. Everything else in the flow is built on top of this one reusable connection.

Service type

HTTP service

Base URL + authentication profile used by http adapter blocks.

Database service

Host, credentials and capabilities used by jdbc adapter blocks.

External Kafka receiver

Input only — external clusters are never a destination.
Currently unavailable

Sink destination

Endpoint + credentials the managed Connect sinks write to — selected by kafka+connect and kc blocks.

Connection details

HTTP service

Configure the fields required by this service type.

Name

Base URL

https://api.example.corp

Auth mode

API key

Key name

X-API-Key

Key location

Header

Key value

API gateway egress

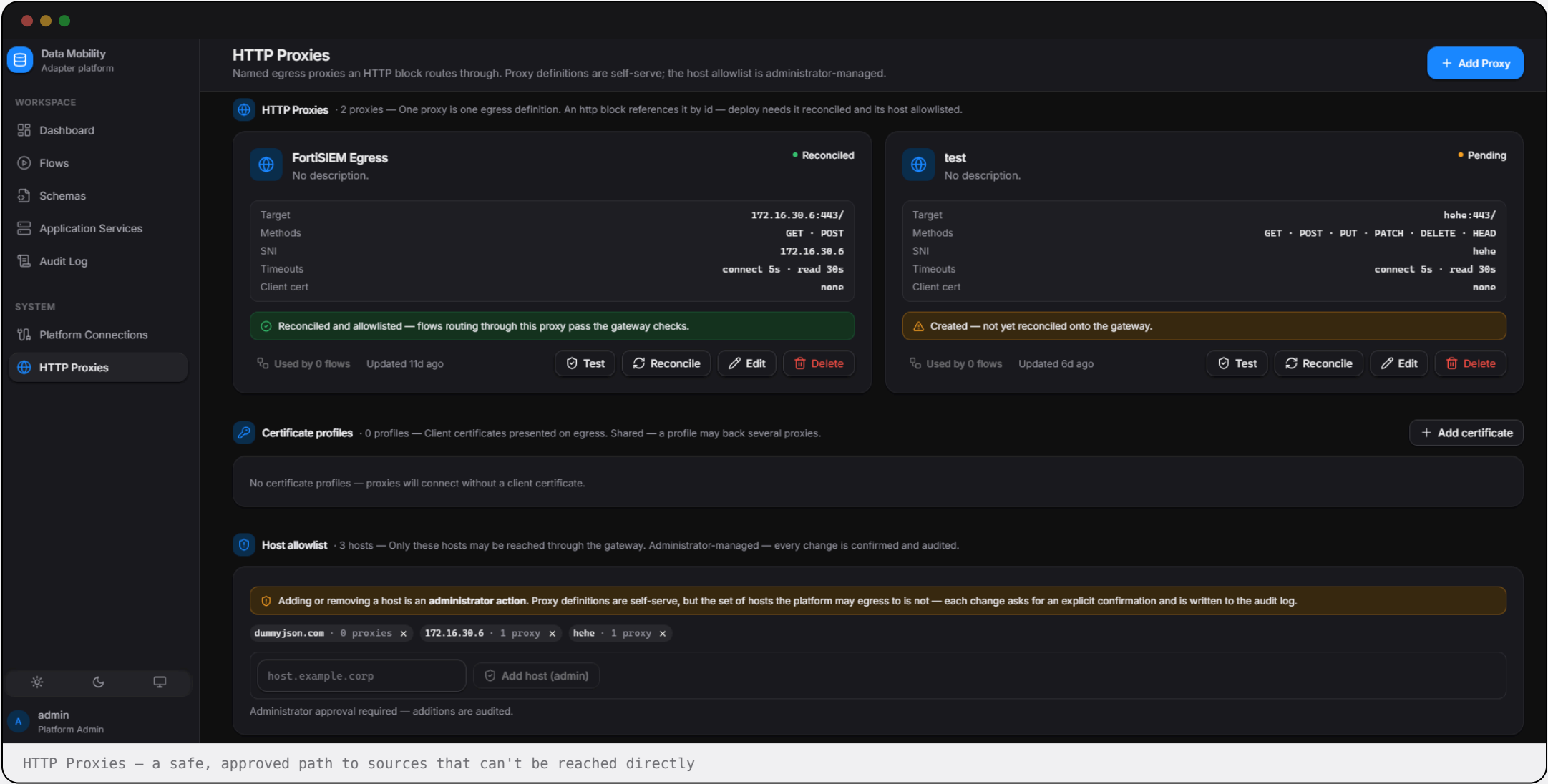
No proxy — call the host directly

STEP · 2 / 8

Route around access issues

ONLY IF NEEDED

If the source can't be reached directly — a missing certificate, a private network, no clean path in — the team defines a proxy instead: a target, allowed methods, timeouts, and an approved host list. The flow routes through it exactly as if the connection were direct.



STEP · 3 / 8

Build the flow

Back in the Flow Builder, the first thing to do is pick the adapter tied to that service — HTTP, JDBC, whatever the source speaks — then lay out the path to Kafka as a visual map. No code, and nothing hidden in a script only one person understands.

Data Mobility

Adapter platform

WORKSPACE

Dashboard

Flows

Schemas

Application Services

Audit Log

SYSTEM

Platform Connections

HTTP Proxies

admin

Platform Admin

Rapid7 Securado - Tags

Tags → tag assets + tag sites, raw JSON to Kafka

Draft

Save

Deploy

More

Adapter configuration

Flow map

Expand

Collapse

http-read

List Tags

http-read

Tag Detail

http-read

Tag Assets

http-read

Tag Sites

Publish Tag

tag

Publish Tag Asset

tag_asset

Publish Tag Site

tag_site

bronze.rapid7_securado_

raw.rapid7_securado_ta_

bronze.rapid7_securado_

raw.rapid7_securado_ta_

bronze.rapid7_securado_

raw.rapid7_securado_ta_

Adapter settings

GET /tags/\${tag_id}/assets

API documentation

Upload an OpenAPI 3.0/3.1 JSON document to pick Method and Path from it below — optional.

Upload

Method

GET

Path

/tags/\${tag_id}/assets

https://apisix.datapasc.com/rapid7_securado/

Response parsing

JSON

\$.resources[*]

split into records

Advanced

page pagination

Headers

Add header

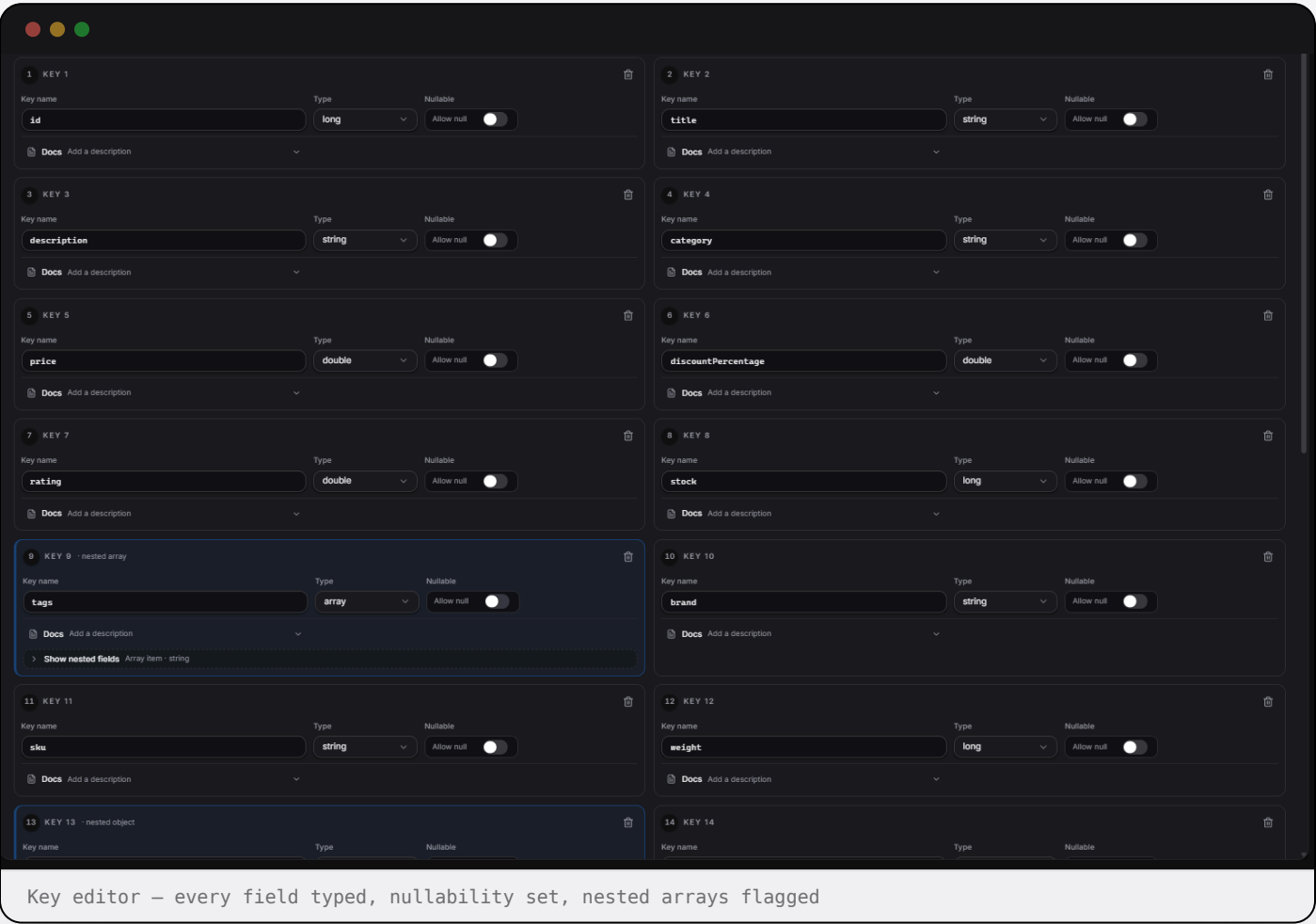
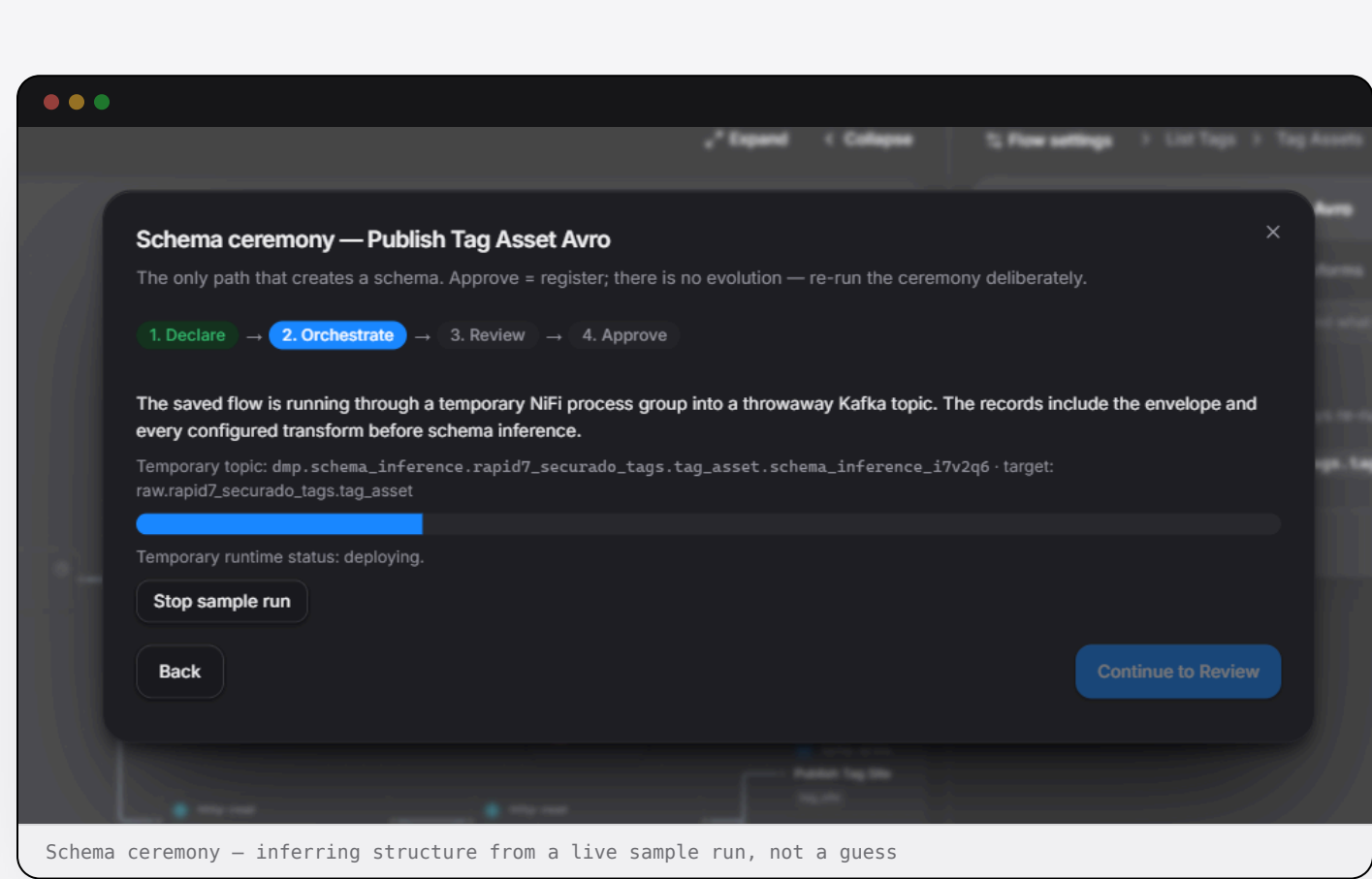
Flow Builder – the pipeline as a map, not a script

STEP · 4 / 8

Become schema-aware

OPTIONAL

Rather than eyeballing field names, the platform runs a real schema ceremony — declare, orchestrate, review, approve — inferring structure from a live sample run before anything is registered. Skip it and publish unvalidated, or let the ceremony produce an approved, versioned schema.



STEP · 5 / 8

Deploy the flow

Deploying pushes the flow into the running NiFi environment — and from that moment, its runtime numbers are the proof: records in, records out, nothing stuck in a dead-letter queue. No greyed-out placeholders, just what's actually moving right now.

Flow operations

Running

Live operational views for this deployed flow

Deployment

State

Enabled

Running

Enabled

Deployed

1h ago

Last run

29m ago

Schedule

cron 0 */6 * * * (UTC)

DLQ topic

dlq.rapid7_securado_site_assets

Entity outputs

kafka · write

Publish Asset

bronze.rapid7_securado.asset

kafka · write

kafka site write

raw.rapid7_securado_site_assets.sites

kafka+connect

Publish Asset Avro

raw.rapid7_securado_site_assets.asset

Schema approved

kafka+connect

Publish Site Avro

raw.rapid7_securado_site_assets.site

Schema approved

kafka · write

Publish Asset Software

bronze.rapid7_securado.asset_software

kafka · write

Publish Asset Service

bronze.rapid7_securado.asset_service

kafka · write

Publish Asset Vulnerability

bronze.rapid7_securado.asset_vulnerability

STEP - 6 / 8 **Send to any destination**

A flow's output isn't locked to one place. The Sink tab defines exactly where its data lands next — the destination is a configuration on the flow, not a rewrite of it.



Rapid7



Kafka



PostgreSQL



HTTP API

Flow settings > List Threats > Threat Timeline > Publish Threat Timeline Avro

kafka+connect Publish Threat Timeline Avro

Identity Adapter Transforms Entity Schema Sink Messages Sink runtime

DESTINATION where the records end up, and what follows

Sink runtime

live connector and task controls

☐ Operational status and actions for this block's Kafka Connect sink. Configuration stays in Sink setup.

Kafka Connect sinks

Per-sink operational control — pause, resume, restart or delete one connector at a time. Configuration lives in the flow builder.

kafka+connect Publish Threat Timeline Avro RUNNING

bronze.sentinelone.threat_timeline_raw.avro_iceberg - org.apache.iceberg.connect.icebergs1nkConnector

task 0 RUNNING kafka-connect:8083

Pause

Stop

Restart

Refresh status

Delete

Deleting takes 1 downstream node with it. Delete block



PostgreSQL



Apache Iceberg



Apache Polaris

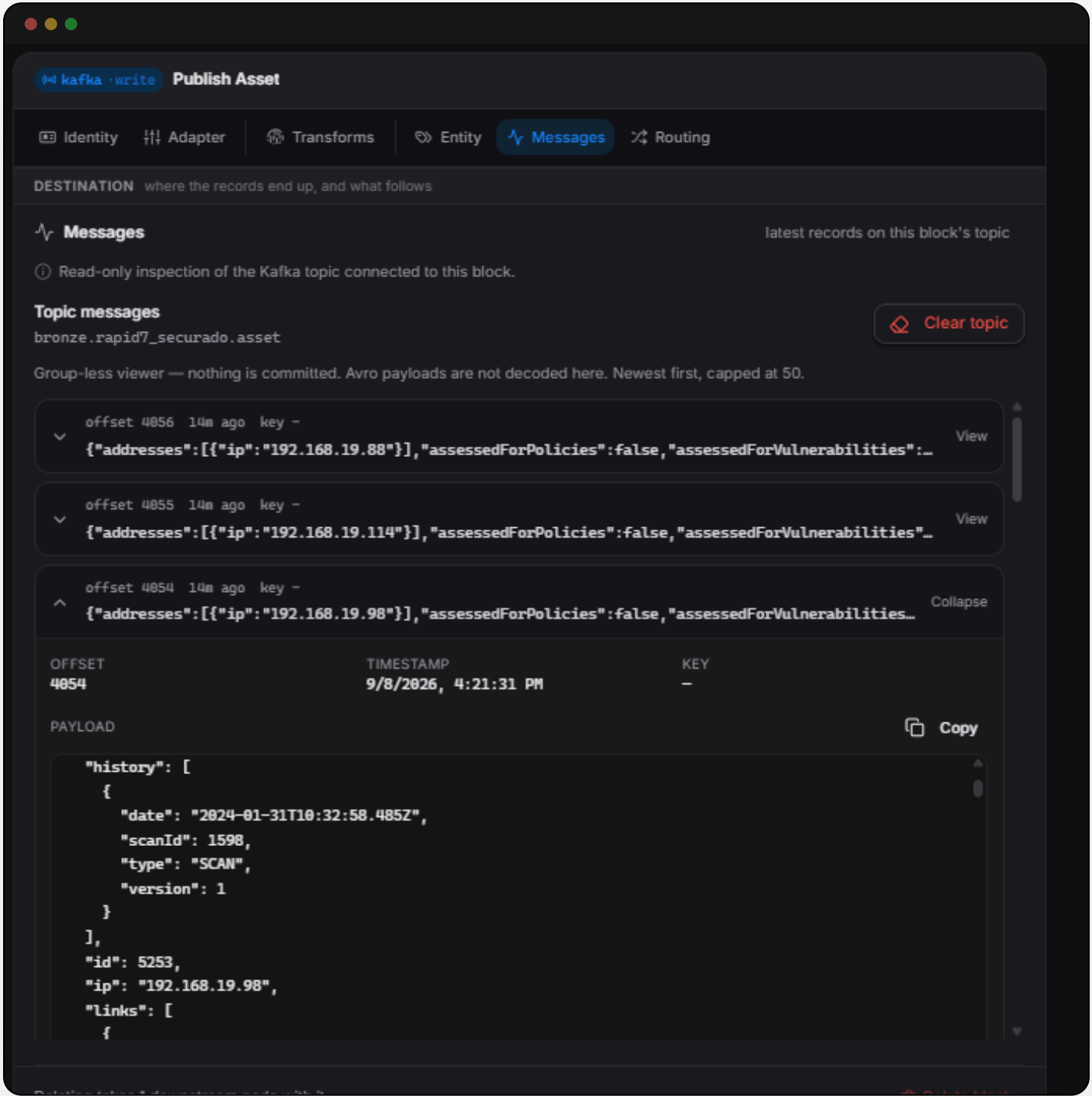


JDBC

STEP · 7 / 8

Verify the data is real

Before trusting a flow, the team watches it work — the real Kafka payloads it's publishing, live, not a delayed report.



STEP · 8 / 8

Monitor your flows

Once a source is live, this is where the team lives day to day — every flow's state, recent activity, and anything that needs attention, in one dashboard. The next source follows the same steps, in a fraction of the time.

72
Flows

0
Running

67
Approved schemas

6/6
Connections healthy

2/61
Sink connectors
44 not deployed

Flow status

Operational state of all flows

HTTP Write Test

ⓘ http · read

Stopped

Kafka Read Test

ⓘ kafka · read

Stopped

SentinelOne - Threats

ⓘ http · read

Stopped

Trino CMDB - asset_groups

ⓘ jdbc · read

Stopped

Codex DummyJSON Cleanup E2E

ⓘ http · read

Draft

DummyJsonTest

ⓘ http · read

Draft

FortiSIEM Agent Status

ⓘ http · read

Draft

FortiSIEM CMDB Catalog

ⓘ http · read

Draft

FortiSIEM Device Inventory

ⓘ http · read

Draft

FortiSIEM Incident

ⓘ http · write

Draft

View all

View all 72 flows (62 more)

Recent activity

Last admin actions

Bulk stop finished

1 succeeded, 0 failed

13s ago · admin

Flow stopped

SentinelOne - Threats

13s ago · admin

Bulk stop started

1 flow(s)

15s ago · admin

Bulk start finished

1 succeeded, 0 failed

10m ago · admin

Flow started

SentinelOne - Threats

10m ago · admin

Bulk start started

1 flow(s)

10m ago · admin

Bulk enable finished

1 succeeded, 0 failed

10m ago · admin

Bulk enable started

1 flow(s)

10m ago · admin

Needs attention

Drafts, drift, and service flags

68

Codex DummyJSON Cleanup E2E

Draft — not yet deployed

DummyJsonTest

Draft — not yet deployed

FortiSIEM Agent Status

Draft — not yet deployed

FortiSIEM CMDB Catalog

Draft — not yet deployed

FortiSIEM Device Inventory

Operations trend

Audit activity over the last 7 days

172

87

29

100

24

Wed

Thu

Fri

Sat

Sun

Mon

Tue

Success

Warning

Failed

Dashboard — every flow, connection, and sink, at a glance

Flow operations

Live operational views for this deployed flow

Running

Records (24h)

0

Errors (24h)

0

Queued

0

Last run

—

BLOCK	IN	OUT	QUEUED
http · List Threats	0	0	0
kafka · Publish Threat	0	0	0
http · Threat Timeline	0	0	0
http · Threat Notes	0	0	0
kafka · Publish Threat Timeline	0	0	0
kafka · Publish Threat Note	0	0	0
kafka_kc · Publish Threat Avro	0	0	0
kafka_kc · Publish Threat Timeline Avro	0	0	0

Topic message counts

bronze.sentinelone_securado.threat

82

bronze.sentinelone_securado.threat_note

0

bronze.sentinelone_securado.threat_timeline

3,854

dlq.sentinelone_threats

0

raw.sentinelone_threats.threat

82

Per-flow metrics — records, errors, and queue depth, per block

Architecture of Data Mobility Platform

How the application orchestrates NiFi, Kafka, Redis, Apicurio, APISIX, and Kafka Connect

